

1-Events of reflex actions include all the following, except :-

- a- stimulation of sensory receptors
- b- stimulation of central synapses
- c- inhibition of sensory receptors
- d- inhibition of central synapses

2-A reflex arc includes:

- a- At least two sets of sequential neurons
- b- At least two sequential sets of central synapses
- c- At least two types of sensory receptors
- d- At least two types of efferent neurons

3- Muscle spindles:

- a- Are found in all skeletal muscles
- b- Are found only in large skeletal muscles
- c- Consist of small numbers of extrafusal muscle fibers
- d- Consist of a large number of extrafusal muscle fibers

4-Stretch of an innervated muscle evokes :-

- a- contraction of its spindles
- b- contraction of its extrafusal fibers
- c- contraction of antagonistic muscles
- d- relaxation of synergistic muscles

5- The nuclear-bag fibers of muscle spindles are innervated by:

- a- C nerve fibers
- b- A β nerve fibers
- c- A δ nerve fibers
- d- Ia nerve fibers

6- The nuclear-chain fibers of spindles are innervated by:

- a- A α and A δ nerve fibers
- b- A δ and C nerve fibers
- c- Ia and II nerve fibers
- d- Only type II nerve fibers

7- The central ends of afferents from muscle spindles synapse with all the following types of neurons, except:

- a- α -motor neurons of the same muscle
- b- γ -motor neurons of the same muscle
- c- 2nd order neurons of ascending sensory pathways

- 8- Gamma-motor innervations of muscle spindles produce:
- a- Contraction of the central region of the spindle fibers
 - b- Increased sensory discharge from the central region of the spindle fibers
 - c- Decreased sensory discharge from the central region of the spindle fibers
 - d- Relaxation of the peripheral regions of the spindle fibers
- 9- Discharge from muscle spindles could be increased by all the following, except:
- a- Increased α -motor neuron discharge
 - b- Increased γ -motor neuron discharge
 - c- Stretch of the intrafusal muscle fibers
 - d- Stretch of the extrafusal muscle fibers
- 10) γ -motor neurons control muscle tone by :-
- a- adjusting the supraspinal facilitatory discharge
 - b- adjusting the α -motor neuron discharge
 - c- adjusting the muscle spindle discharge
 - d- adjusting the activity of interneurons in the reflex arc of muscle tone
- 11-The role of γ -motor neurons in regulation of equilibrium is achieved by :-
- a- initiating contraction of spindle fibers which directly antagonize postural deviation
 - b- increasing spindle sensitivity to stretch
 - c- directly stimulating extrafusal muscle fibers to antagonize postural deviation
 - d- directly adjusting the discharge of α -motor neurons innervating the extrafusal muscle fibers to antagonize postural deviation
- 12- Increased γ -motor neuron discharge stimulates muscle spindles, because it:
- a- Produces stretch of the extrafusal muscle fibers
 - b- Causes stretch of the peripheral regions of the intrafusal fibers
 - c- Causes stretch of the central region of the intrafusal fibers
 - d- Stimulates directly the sensory fibers innervating muscle spindles
- 13- The gamma efferent neurons:
- a- Controls spindle activity during voluntary movements.
 - b- Supplies the central zone of intrafusal muscle fibres.
 - c- Is inhibited by the vestibule-spinal tract.
 - d- When stimulated, causes less discharge of muscle spindles.
- 14- Gamma-motor neurons control muscle tone by adjusting the:
- a- Supraspinal facilitatory discharge
 - b- α -motor neuron discharge
 - c- Muscle spindle discharge
 - d- Activity of interneurons in the reflex arc of muscle

15- Co-activation of α and γ -motor neurons:

- a- Increases γ -motor neuron discharge whenever the activity of α -motor neurons increases
- b- Is mediated by interneurons that link the α and γ -motor neurons
- c- Maintains the proprioceptive information to higher centers during muscle contraction
- d- Increases the α -motor neuron discharge whenever the activity of γ -motor neurons increases

16- Whenever the position of a joint is stabilized at a certain attitude, the nervous system produces this by:

- a- Increasing the α -motor neuron discharge to all muscles attached to the joint
- b- Increasing the γ -motor neuron discharge to all muscles attached to the joint
- c- Increasing γ -motor neuron discharge to postural muscles
- d- Co-activation of α and γ -motor neurons innervating the involved muscles

17- The shortest reflex time is recorded with:

- a- A flexor withdrawal reflex
- b- An inverse stretch reflex
- c- A stretch reflex
- d- A scratch reflex

19-stretch reflex is characterized by the following except :-

- a- disynaptic reflex
- b- high localization.
- c- it is of graded response

20- When a skeletal muscle is suddenly stretched it:

- a- Relaxes suddenly
- b- Develops a static stretch reflex
- c- Develops a dynamic stretch reflex
- d- Develops clonic contractions

21- A tendon jerk is:

- a- A dynamic stretch reflex
- b- A static stretch reflex
- c- Evoked by gradually stretching the muscle
- d- Evoked by stimulation of tendon receptors

22) interruption of γ -motor neuron discharge to a skeletal muscle produces :-

- a- contraction of the muscle
- b- contraction of the spindle fibers
- c- relaxation of the spindle fibers
- d- increased sensory discharge from the spindle fibers

23-The role of muscle spindles in the maintenance of the upright posture depends on all of the following, except :-

- a- contraction of the peripheral contractile part of spindle fibers
- b- increased sensory discharge from spindles of postural muscles
- c- increased supraspinal facilitation to the γ -motor neurons of postural muscles
- d- presence of greater numbers of spindles in postural muscles

24-When the γ -motor neuron discharge to a muscle decreases, it causes :-

- a- increased muscle spindle discharge
- b- shortening of the spindle fibers
- c- stronger contraction of the extrafusal muscle fibers
- d- decreased spindle sensitivity to stretch

25- Antigravity muscles maintain stretch reflex for prolonged periods without fatigue, because they:

- a- Are heavily innervated by α -motor neurons
- b- Obtain their energy needs mainly from anaerobic metabolic processes
- c- Contain exceptionally high levels of creatine phosphate
- d- Are rich in mitochondria

26- Skeletal muscle tone is:

- a- A dynamic stretch reflex
- b- A state of alternative contraction and relaxation
- c- Increased during rest
- d- Decreased during standing upright

27-Interruption of spindle discharge from a muscle causes contraction of the muscle to become jerky and irregular due to :-

- a- increased activity of reverberating circuits causing fluctuation of the motor discharge to the muscle
- b- increased activity of inhibitory interneurons causing oscillating inhibition of the α -motor neurons of the muscle
- c- irregular discharge of excitatory inputs to the α -motor neurons of the muscle
- d- post-tetanic potentiation of the α -motor neurons of the muscle

28- Which of the following is a supraspinal excitatory to the gamma motor neurons?

- a- Medullary reticular formation
- b- Red nucleus.
- c- Neocerebellum
- d- Paleocerebellum.
- e- Motor area 6

29- All the following about muscle tone is true, except:

- a- Assessed as a resistance offered by a muscle on a passive stretch.
- b- Reduced during sleep
- c- Altered during stress.
- d- Increased after loss of function of motor area 4
- e- More in antigravity muscle

30- Tendon jerks are clinically examined to asses:

- a- Integrity of muscle spindles
- b- Integrity of reflex pathway
- c- The total reflex time of the jerk
- d- Central delay time of the jerk

31- Absence of a tendon jerk could result from any of the following conditions, except:

- a- Lesions of supraspinal facilitatory centers
- b- Lesions of the efferent neurons
- c- Lesions of the afferent neurons
- d- Lesions of the spinal nerve centers

32- Exaggeration of tendon jerks could result from any of the following conditions, except:

- a- Lesions of supraspinal facilitatory centers
- b- Lesions of supraspinal inhibitory centers
- c- Increased γ -motor neuron discharge
- d- Anxiety

33- Decerebrate rigidity is due to:

- a- Stimulation of type I b sensory neurons
- b- Loss of cerebellar inputs to the red nucleus
- c- Overactivity of the medullary reticular nuclei involved in motor control
- d- Unopposed activity of the pontine reticular nuclei
- e- Degeneration of the nigrostriatal pathway

34- Activity of RAS produces:

- a- Sleep
- b- Awakening
- c- Drowsiness.
- d- Coma

35- Decreased muscle tone occurs in:

- a- Activation of γ -fibers
- b- Anxiety
- c- LMNL
- d- Parkinson's disease
- e- UMNL

36- The discharge from Golgi tendon organs initiated by excessive stretch of a skeletal muscle produces:

- a- Inhibition of α -motor neurons of antagonistic muscles
- b- Inhibition of γ -motor neurons of antagonistic muscles
- c- Inhibition of α -motor neurons of the same muscle
- d- Stimulation of γ -motor neurons of the same muscle

37- Regarding the role of muscle spindles and Golgi tendon organs:

- a- Muscle spindles measure the tension of skeletal muscle.
- b- Golgi tendon organs measure the length of skeletal muscles.
- c- Muscle spindles receive both afferent and efferent nerve fibers
- d- Muscle spindles are only of importance in the regulation of posture.

38- Which of the following reflexes is correctly paired with its receptor?

- a- Autogenic inhibition – muscle spindle
- b- Reciprocal inhibition – Golgi tendon organ
- c- Reciprocal inhibition – Pacinian corpuscle
- d- Stretch reflex – muscle spindle
- e- Golgi tendon reflex – Meissner corpuscle

39- On overstretching a normally innervated skeletal muscle, it relaxes due to:

- a- Decreased discharge from the annul spiral endings.
- b- Decreased discharge in the gamma efferent neurons.
- c- Increased discharge from the antagonistic muscle.
- d- Increased activity of the alpha motor neurons.
- e- Increased activity of the Golgi tendon organ.

40- Inverse stretch reflex:

- a- Increases the possibility of avulsion of the excessively stretched muscle from its bony attachments
- b- Is clinically manifested by lengthening reaction
- c- Is clinically tested by examining the tendon jerks

41- γ -motor neuron discharge to a muscle is inhibited by impulses reaching the γ -motor neurons from all the following source, except :-

- a- Golgi tendon organs of the same muscle
- b- spindles of antagonistic muscles
- c- medullary reticular formation
- d- pontine reticular formation

42- Clonus:

- a- Is a sign of decreased supraspinal facilitation
- b- Is manifested as oscillating mechanical vibrations following tendon jerks
- c- Associates exaggeration of tendon jerk

43. Concerning the flexor withdrawal reflexes:

- a- They involve reciprocal innervations
- b- Their central delay equals 0.5 msec.
- c- It is a monosynaptic reflex.
- d- It provides an example of a deep reflex.

44- The crossed extensor reflex:

- a- Provides a typical example of a deep reflex.
- b- Is monosynaptic.
- c- Does not involve reciprocal inhibition.
- d- Supports the body against gravity.

45- Regarding reflexes:

- a- A monosynaptic reflex arc involves one or more interneurons
- b- The knee jerk reflex is an example of a stretch reflex
- c- Interneurons are the final common path for all reflexes
- d- Withdrawal reflexes are lost following cervical section of the spinal cord.

46- Spinal shock is due to:

- a- Severe pain felt at the site of the lesion
- b- Severe hypotensive shock
- c- Interruption of the ascending sensory pathways
- d- Interruption of the descending facilitatory tracts

47- The stage of spinal shock is characterized by:

- a- Loss of sensations from the body
- b- Paralysis of all skeletal muscles innervated by the spinal cord
- c- Exaggerated tendon jerks
- d- Failure of spinal reflexes below the level of the lesion

48- In humans the usual duration of the stage of spinal shock is:

- a- 2-6 hours
- b- 2-6 days
- c- 2-6 weeks
- d- 2-6 months

49- Failure of spinal reflexes during the stage of spinal shock causes:

- a- Automatic micturition
- b- Hypotension
- c- Babinski sign
- d- Spasticity of the paralyzed muscles

50- Complete transection of the spinal cord produces all of the following effects, except:

- a- Permanent loss of all sensations mediated by the cord below level of lesion
- b- Permanent loss of voluntary movements by muscles innervated by the cord below level of lesion
- c- Permanent loss of reflexes mediated by the cord below level of lesion
- d- Temporary loss of micturition reflexes

51- Long term consequence of transection of the spinal cord at T12 level includes:

- a- Paralysis of the bladder muscle and loss of the ability to defecate
- b- Severe drop of ABP
- c- Loss of the reflexes concerned with erection and ejaculation
- d- flexor spasm on stimulation of the inner aspect of the thigh

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52- At the beginning of head rotation in a horizontal plane to the left:
a- Endolymph filling the S.C.C moves bending the cristae of both sides to the right
b- Stereocilia of the hair cells of the left crista hyperpolarized
c- Stereocilia of the hair cells of the right crista depolarized
d- Firing rate on the right vestibular nerve increases

53- The otolith organs (utricle and saccule) are responsible for:
a. Producing the vestibular-ocular reflex
b. Detecting the position of the head in space
c. Producing rotary nystagmus
d. Detecting angular acceleration
e. Producing the stretch reflex

54- Which of the following is maximally sensitive to linear head movements in the vertical plane?
a- Macula of the utricle
b- Macula of the saccule
c- Crista ampullaris of the anterior semicircular canal
d- Crista ampullaris of the horizontal semicircular canal

55- The otolith organs are receptors for:
a- Angular acceleration
b- Linear acceleration
c- Vibration sense.
d- Proprioceptive.

56- All are true about the vestibular apparatus, except:
a- Bony labyrinth contain 3 SCCs.
b- Helps to maintain equilibrium.
c- Is stimulated only during linear acceleration.
d- Is stimulated during linear and angular acceleration.

57- Regarding the vestibular function:
a- The SCCs do not sense angular movement at a slow speed.
b- The utricle records the position of the head as well as linear acceleration.
c- Is the only sensory input required for the maintenance of body equilibrium
d- Damage to the vestibular apparatus causes positive Romberg's sign

58- When a person starts to rotate towards the right, all is true, except:

- a- The stereocilia of the hair cells of the right horizontal canal bend towards the kinocilium.
- b- Both eyes move slowly to the left.
- c- Hair cells of the crista of the right SCCs depolarize.
- d- The endolymph in the canals moves in direction of rotation.

59- Nystagmus:

- a- Is observed only at the beginning of rotation.
- b- Has no aim.
- c- Is not observed during rotation.
- d- Slow component, is in the opposite direction of rotation.

60- Post-rotating effects:

- a- Are cortical voluntary movements of the limbs and body.
- b- Aim to correct false sensation of rotation.
- c- Aim to prevent falling of the body.
- d- all of the above.